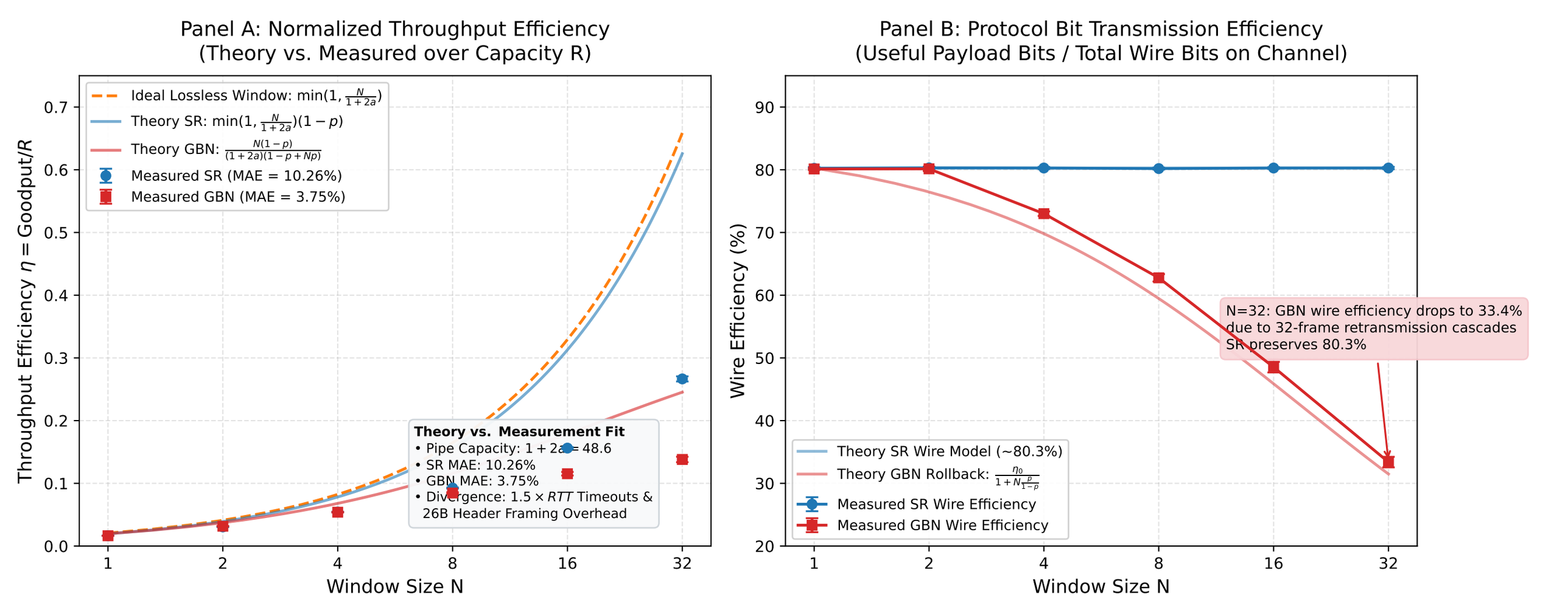
Panel A: Normalized Throughput Efficiency
(Theory vs. Measured over Capacity R)
Ideal Lossless Window: min(1, $\frac{N}{1+2a}$)
Theory SR: min(1, $\frac{N}{1+2a}$)(1 − p)
Theory GBN: $\frac{N(1-p)}{(1+2a)(1-p+Np)}$
Measured SR (MAE = 10.26%)
Measured GBN (MAE = 3.75%)
Theory vs. Measurement Fit
• Pipe Capacity: 1 + 2a = 48.6
• SR MAE: 10.26%
• GBN MAE: 3.75%
• Divergence: 1.5 × RTT Timeouts &
26B Header Framing Overhead
Throughput Efficiency η = Goodput/R
Window Size N
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
1
2
4
8
16
32
Panel B: Protocol Bit Transmission Efficiency
(Useful Payload Bits / Total Wire Bits on Channel)
Theory SR Wire Model (~80.3%)
Theory GBN Rollback: $\frac{\eta_0}{1+N\frac{p}{1-p}}$
Measured SR Wire Efficiency
Measured GBN Wire Efficiency
N=32: GBN wire efficiency drops to 33.4%
due to 32-frame retransmission cascades
SR preserves 80.3%
Wire Efficiency (%)
Window Size N
20
30
40
50
60
70
80
90
1
2
4
8
16
32